ITA06020 MACHINE LEARNING

Experiment 18 – Perceptron-Based Iris Flower Classification

Aim:
To classify Iris flowers into different species using the Perceptron algorithm.

Procedure:

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing data.
4. Standardize the input features.
5. Train the Perceptron model.
6. Predict flower species and calculate accuracy.

Dataset: Iris Flower Dataset (built-in Scikit-learn dataset).

Python Code:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.linear_model import Perceptron

from sklearn.pipeline import make_pipeline

from sklearn.metrics import accuracy_score, confusion_matrix

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(

   X, y, test_size=0.2, random_state=42, stratify=y

)

model = make_pipeline(

   StandardScaler(),

   Perceptron(max_iter=1000, random_state=42)

)

model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred) * 100, "%

sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(sample)

print("Predicted Flower:", iris.target_names[prediction[0]])

print("Confusion Matrix:")

print(confusion_matrix(y_test, y_pred))
```

Output:

```
Accuracy: 86.66666666666667 %
Predicted Flower: setosa
Confusion Matrix:
[[10  0  0]
 [ 0  7  3]
 [ 0  1  9]]
```

Result:

The Iris flower classification was successfully performed using the Perceptron algorithm with an accuracy of 86.67%.